

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-12 13:00 UTC

SEC P vs NP — notebook summary

Generated 2026-05-12 13:00 UTC

- Cycles recorded: **658**
- Time span: 447.3h (~1.47 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	621
BARRIER_HIT	18
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	36
Schur-Weyl Duality	23
Representation Theory of Symmetric Groups	17
Matroid Theory	16
Free Probability	14
Additive Combinatorics	10
Free Probability Theory	10
Noncommutative L^p Geometry	9
Random Matrix Theory	8
Polymatroid Theory	8
Algebraic Geometry	7
Spectral Graph Theory	7
Finite Geometry	6
Plethysm Theory	6
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4

Field	Cycles
Persistent Homology	4
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Noncommutative Harmonic Analysis	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-12 06:36 UTC	INCONCLUSIVE	Invariant Ring Degree Bounds SOS Refutation Degree for Symmetric
2026-05-12 06:43 UTC	INCONCLUSIVE	Noncommutative L^p Norm of Communication Matrix Bounds Randomized
2026-05-12 07:48 UTC	INCONCLUSIVE	Matroid Rank Inverse Proportional to Monotone DNF Size for k-CLIQ
2026-05-12 07:54 UTC	INCONCLUSIVE	Matroid Rank Submodularity in Monotone DNF for k-CLIQUE
2026-05-12 08:01 UTC	INCONCLUSIVE	Tensor Product Multiplicity Inverse Proportional to Monotone Circ
2026-05-12 08:14 UTC	INCONCLUSIVE	Persistent Homology of Read-Twice BPs Bounded by Log-Size
2026-05-12 08:22 UTC	INCONCLUSIVE	Hilbert Function Growth Inversely Proportional to Extended Frege
2026-05-12 08:32 UTC	INCONCLUSIVE	Block Design Discrepancy and Communication Complexity Lower Bound
2026-05-12 08:44 UTC	INCONCLUSIVE	Second Eigenvalue Inverse Proportional to Resolution Proof Length
2026-05-12 09:26 UTC	INCONCLUSIVE	Real Radical Dimension Bounds SOS Degree for Max-CUT
2026-05-12 10:22 UTC	INCONCLUSIVE	Noncommutative Fourier Coefficient Sum Inversely Proportional to

Time	Verdict	Title
2026-05-12 10:30 UTC	INCONCLUSIVE	SOS Degree Lower Bound via Eigenvalue Count in Moment Matrix
2026-05-12 11:18 UTC	INCONCLUSIVE	Kronecker Coefficient Gap in Symmetric Decompositions of Permanen
2026-05-12 11:48 UTC	INCONCLUSIVE	Schur-Weyl Decomposition Component Count
2026-05-12 12:40 UTC	INCONCLUSIVE	vs. Monotone Circuit Siz Real Radical Degree Lower Bound for AC^0 Circuits Computing PARITY

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
  pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```